# Exposure Concentration Under a Learned Ranker Is a Cold-Start Effect: A Convergence Test in a Simulated Matching Market

**Abstract**

A recommender that decides which candidates a matching market's agents ever see changes what a preference list means, because a preference over a truncated list is not the preference that operates over the whole population. This paper measures how far that goes and how long it lasts. A finite two-sided market of 1000 agents carries a four-dimensional trait vector and a private commitment type; a neural collaborative filter produces top-20 shortlists; truncated deferred acceptance clears the market over those alone. Exposure Gini under the learned ranker exceeds an oracle's by $+0.0251$ $[+0.0155, +0.0347]$ over five seeds at a hundred epochs. Extending to three hundred shows that is not a floor: the gap peaks at $+0.3055$ and is below 0.001 from about epoch 250, where it stops being distinguishable from flat. That is a bounded residual, since a flat tail is also what a very slow decay looks like. The ordering is positive in all twelve cells of a parameter sweep, but its size varies fivefold, and the attribution to the filter learning coverage fails in the one cell where coverage never climbs, so the mechanism holds at one point, not generally. A companion institutional comparison yields an interior welfare optimum. Five seeds carry every headline and no parameter is calibrated.

## 1 Introduction

When a recommender sits between the two sides of a matching market, it decides which candidates anyone ever sees. The consequence is easy to state and awkward to price: a preference expressed over a truncated list is not the same preference that operates over the whole population. The agent on nobody's list is not rejected; she is never considered, and no preference data will record the difference. This paper asks what happens to a matching market under that arrangement, and how it compares with a differently designed market that trades liquidity for information.

For a single scalar notion of status the aggregate questions have closed-form answers, and they have recently been settled. When unions must be hypergamous, that is, when a partner must rank at or above rather than beside, the minimal unmatched mass is a functional of the two marginal status distributions alone, and no behavioural strategy changes it. Two things the literature treated as substantive, the feasibility of a status requirement and the size of the resulting deficit, turn out to be properties of histograms, and any new work in that setting is at best a check on a theorem.

This paper works where that theorem does not reach. The constraint here is *visibility*, not compatibility: the feasible set is the top-$k$ of an estimated multi-attribute utility, lists differ by agent, and estimator error is correlated in a way no marginal captures. The closed-form result relies on the constraint being a nested upper set, which is exactly what makes the deficit a one-dimensional supremum. Truncated lists from a learned ranker are not nested, and their error depends on how long the ranker has trained. The structure that makes the deficit tractable is absent rather than merely harder, so the question must be answered numerically.

The answer depends on *when* you ask. That is the paper's central finding and its central methodological warning.

## What was done

A finite two-sided market of a thousand agents carries a four-dimensional trait vector and a private commitment type. A neural collaborative filter trained on the swipe log produces a top-20 shortlist per agent, and a truncated deferred-acceptance pass clears the market over those shortlists and nothing else. In parallel, a second institution is modelled as a subgraph of the same population, with complete visibility inside it and a positive signalling cost: information instead of liquidity, where the global market has liquidity instead of information. Welfare is scored on a common basis, so agents the subgraph excludes are priced at their counterfactual outcome rather than deleted, since otherwise an entry filter improves its own arithmetic by removing the agents it cannot serve.

Six hypotheses were posed in advance, each with a stated kill condition, and four kinds of test applied: seed-bootstrapped intervals, a horizon extended to convergence, a twelve-cell parameter sweep with its claim thresholds fixed before the cells were read, and multiplicity correction over the forty institutional cells the headline optimum was selected from. §4 states the sample sizes so the numbers can be checked.

## What was found

**Estimation error concentrates exposure, and the concentration is a transient.** The learned ranker's exposure Gini exceeds an oracle's by $+0.0251$ at the shipped horizon, positive on all five seeds. Extending the run to 300 epochs shows this is not a floor but a decay: the gap peaks shortly after the start, is $+0.0241$ at epoch 99, and is below 0.001 from about epoch 250, where it stops being distinguishable from flat at five seeds. That is a bounded residual rather than a demonstrated limit, because a flat tail is also what a very slow decay looks like, and this design cannot separate the two. A hundred-epoch measurement of this quantity describes a transient, and every number quoted at that horizon carries it.

**The concentration is real everywhere, but its size is not constant.** The gap is positive with an interval excluding zero in all twelve sweep cells and reverses nowhere, while its magnitude varies fivefold. A reader should quote the range with its condition, or the shipped cell with its regime, never one number as "the" gap.

**The explanation is conditional.** The mechanism offered, the filter learning whom it can represent, holds where coverage climbs and fails in the one cell where it does not. There the gap is also small, but because both arms concentrate rather than because the ranker caught up. Convergence to zero is established at one point in parameter space, not as a general property of mediation.

**An interior institutional optimum exists, and it is narrow.** At a signalling cost low enough that entry screens little, welfare under the subgraph institution exceeds the global oracle's by $+64.62$ at subgraph size 0.65, and survives correction for the forty cells it was chosen from. Only two cells survive that correction, so the optimum is a two-cell feature rather than a region.

Three of six hypotheses are qualified. One is untestable because the mechanism it names is absent from the shipped rule, one holds at a single cost level, and one cannot be tested at all because the model has no attrition margin. They are reported at the same length as the results that worked, because a paper that reports only its successes reads as advocacy.

### Why it matters, and what it does not claim

That mediation's effect is horizon-dependent is absent from the recommender-loop literature, which predicts homogenisation that persists or compounds, and from the closed-form literature, which has no horizon argument. Had the effect been measured only at the shipped horizon, or at twenty epochs as an earlier version of this work was, the paper would have reported a standing deficiency of learned rankers where the truth is a start-up cost. The general lesson is that a single-horizon inequality cannot distinguish a standing bias from a transient, and that the distinction is measurable.

Nothing here is a claim about a population. Every parameter is a modelling choice, nothing is calibrated, five seeds carry every headline, and one population size carries it. The results bound what this mechanism can do under a stated utility; §6 says so at length rather than in a footnote.

## 2 Related work

### 2.1 The closed-form result, and what it settles

For a marriage market where status is a single scalar and unions must be hypergamous, the aggregate questions have closed-form answers. Bilderbeek [2026] proves the picture: universal hypergamy is feasible exactly when one side's status distribution first-order stochastically dominates the other's, and when it is not, the minimal unmatched mass is the hypergamy deficit $D = \sup_t (G_W(t) - G_M(t))_+$, by a continuum Hall marriage theorem. No behavioural strategy changes either fact, status gains are transfers of rank rather than creations of it, and under status-ranked preferences the stable matching is unique and positive assortative, the Gale–Shapley result of Gale and Shapley [1962] on the continuum.

The consequences for anyone modelling this are severe. Unmatched mass is a functional of the two marginal status distributions alone, so interventions that rearrange who meets whom without changing those marginals cannot move it. This paper does not re-derive the result; it reproduces it against the simulator (§4.5) and then changes the constraint.

### 2.2 The empirical record, and why a model is the instrument

The empirical literature is consistent and mostly about marginals. Becker [1973] supplies the complementarity condition: supermodular surplus implies positive assortative matching, so measured sorting is evidence about the surplus function. Schwartz and Mare [2005] trace US educational homogamy from 1940 to 2003, falling then rising. Eika et al. [2019] find assortative mating accounts for a non-negligible share of cross-sectional household income inequality while *changes* in assortative mating barely move trends: a level effect without a trend effect, and a warning about which one a cross-sectional model can speak to. Esteve et al. [2016] show the educational gender-gap reversal shifting marriage patterns toward hypogamy with no preference change required.

On the micro side, Fisman et al. [2006] randomise speed-dating encounters and find trait weights differing by sex and selectivity rising with group size; Lin and Lundquist [2013] show trait salience is not separable, which bounds the additive utility used here. In online settings, Hitsch et al. [2010] estimate mate preferences, feed them into deferred acceptance, and find predicted stable matches track observed ones closely, concluding that preferences rather than frictions explain most sorting. Bruch and Newman [2018] find a desirability hierarchy consistent across cities, with reply probability falling sharply in the desirability gap.

Why model rather than estimate? Because every instrument above observes *realised* matches inside one institution, and none can answer a counterfactual about a different one. The question

here is institutional: what happens when the list an agent sees is produced by an engagement-optimising estimator rather than given. Kanoria and Saban [2021] and Halaburda et al. [2018] take the same view from the design side, deriving platform welfare from a model of restricted choice rather than estimating it. Galichon and Salanié [2022] make the structural position precise for identification, which is why this paper reports welfare only inside the model. The precedent for computing when the closed form does not apply is the clearinghouse work of Roth and Peranson [1999].

### 2.3 Rankers, feedback loops, and the visibility constraint

He et al. [2017] establish the collaborative-filtering architecture used here and make no claim about markets. Pizzato et al. [2010] introduce the reciprocal recommender and show that accounting for both sides lifts top-ten success substantially; Tomita et al. [2022] describe deploying a matching-theory-based recommender on a live platform. Reciprocity-aware ranking is solved engineering, and a paper claiming novelty for its ranker would be claiming novelty for something built in 2010. Chen et al. [2023] go further, running a matching-model-based ranker on a live platform and measuring realised matches. That is the strongest constraint on any novelty claim here: same intersection, different question.

The loop literature is more directly adversarial. Chaney et al. [2018] show that training on data confounded by the system's own recommendations homogenises users without raising utility. Mansoury et al. [2020] show a loop amplifying popularity bias and hitting minority-group users hardest. Jiang et al. [2019] separate the echo chamber from the filter bubble. Abdollahpouri [2019] frames popularity bias as a fairness problem, and Sonboli et al. [2022] show that in multisided settings fairness criteria for different stakeholders conflict.

Every one of these predicts a loop that *persists or compounds*. The result here points the other way, and the explanation is that the ranker's deficiency is coverage, which training fixes, rather than preference distortion, which it does not. That distinction is not in the loop literature and is the claim a reader should check hardest, which is why §5.8 tests it across the parameter grid and reports the cell where it fails.

Two further results bound what mediation can be credited with. Roth and Rothblum [1999] show truncating a preference list changes outcomes strategically, so a top-$k$ list is not a neutral convenience. Ashlagi et al. [2017] show that in an unbalanced random market even a slight imbalance produces an essentially unique stable matching with the long side matched near-randomly, so some measured asymmetry is competition rather than curation. Both cautions apply: the truncation here is imposed rather than chosen, and the market is balanced, so the competition channel is held at zero by design rather than measured away.

### 2.4 Institutional design and the selection problem

Kanoria and Saban [2021] show a platform that restricts agents' actions can raise welfare. Halaburda et al. [2018] show a platform can compete by limiting the choices it offers, because restricting one agent's choice also restricts her competitors'. Hoppe et al. [2009] give the sharpest constraint on the interior optimum sought here: with two-sided incomplete information the output gain from assortative matching can be exactly offset by signalling costs, which is what failure at high signalling cost looks like.

That literature also names the objection this paper had to design around. If an institution excludes agents, its measured welfare improves for the arithmetic reason that excluded agents are no longer counted. The comparability rule of §3 is therefore a response to a known objection, not

a safeguard invented here, and both accountings are reported so the artefact is visible.

Two results bound any $W$-side number. Roth [1982] establishes that no procedure is both always stable and always strategy-proof, with the proposer-optimal matching simultaneously the acceptor-pessimal one; Knuth [1996] supplies the lattice structure that makes those one statement. Since hypergamy lives in the acceptor's utility, this bias lands where the paper's question is, and every acceptor-side level reported is a lower bound across stable matchings.

### 2.5 What is new here

Thirty-five sources were examined for whether any already does recommender-mediated matching on multiple traits. Thirty-three do not. Three require the claim restated: Chen et al. [2023], Pizzato et al. [2010] and Tomita et al. [2022] between them rule out any novelty claim resting on the ranker. The gap that survives is narrower than "recommender-mediated matching" and is a conjunction of three specific absences: no characterisation of unmatched mass against the visibility constraint, no institutional comparison on a common welfare basis, and no horizon. What this paper adds is the recommender-intermediated setting in which visibility is a truncated top-$k$ list over an estimated multi-attribute utility and no closed form is known. It claims nothing for its ranker, and no progress on the multidimensional-status, search-friction or endogenous-status problems the closed-form literature names as open.

### A note on verification

Reproducing a closed form is verification: it establishes that the machinery is correct, not that the model describes anything. Windrum et al. [2007] and Fagiolo et al. [2007] set out why empirical validation of an agent-based model is a family of practices rather than a single test, and why calibration alone does not resolve it. This paper claims verification and declines the validation question.

## 3 Model

The model is a finite two-sided market cleared once per epoch, with a learned ranker standing between the two sides. Three layers run in a loop: agents form preferences, a collaborative filter decides what each agent can see, a matching algorithm clears the market over what was visible, and the clearing outcome feeds back into the acceptance threshold that gates the next epoch's visibility. This section describes each layer, the two institutional environments the paper compares, the utility that drives them, and the condition under which the feedback channel is live rather than decorative.

### 3.1 The agent layer

Agents are split into a proposer side $M$ and an acceptor side $W$ of equal size. Each carries a four-dimensional state vector $[s, a, c, t]$: status, attractiveness, sociability and age. Status is drawn lognormal; the other three are clipped to the unit interval. Every trait is $z$-scored against population statistics before it reaches a utility, so each weight below is expressed against a unit-variance trait and the weights are directly comparable to one another. The choice to $z$-score is load-bearing for interpretation: it means only the *rank order* of a trait within the population matters, and a distribution with no variance would annihilate its term.

Two further agent attributes drive the institutional comparison. A private *intent* type $\tau_i \in (0, 1]$ is each agent's willingness to commit, invisible to others. A pairwise *alignment* $g_{ij} \in [0, 1]$ measures value compatibility and enters the dissolution hazard rather than the utility.

### 3.2 The recommender layer

A neural collaborative filter with a generalised matrix factorisation path and an MLP path is trained on the swipe log, the record of which pairs were liked, and scores every unseen pair. Each agent then keeps a top-$k$ shortlist of length $k = 20$ from those scores, and *nothing outside that shortlist is visible to the matching step.* This is the architectural commitment the paper rests on: the filter is not a convenience wrapper around a preference list, it *is* the preference list. It is also why a preference expressed over this list is not the preference the same agent holds over the population, which is the point of departure for the whole exercise.

The ranker is deliberately off-the-shelf. It is cited for the estimator and not for a result about markets, and the contribution would be void if it depended on a novel architecture.

### 3.3 The matching layer

Clearing is a truncated deferred-acceptance pass over the shortlists, returning the matched pairs and a blocking-pair count. Truncation matters: with lists truncated at $k$, the procedure is the classical algorithm run on top-$k$ lists rather than on full lists [Gale and Shapley, 1962], and the classical stability guarantee does not carry over unchanged [Roth and Rothblum, 1999]. It also means the clearing bias is a property of the procedure class and not of the algorithm specifically, as the wage-forming counterpart of the same procedure shows [Kelso and Crawford, 1982]. A full-information arm ranks by true utility under the same proposer direction and the same shortlist length, and is the benchmark against which algorithmic loss is measured.

Two properties of the clearing step bound what any $W$-side level can mean. Deferred acceptance with $M$ proposing returns the proposer-optimal matching, which is simultaneously the acceptor-pessimal one: among all stable matchings, every acceptor receives her worst [Roth, 1982, Knuth, 1996]. Since hypergamy lives in the acceptor's utility, this bias lands on exactly the side the paper is about. The full-information and mediated arms use the same direction, so the *gap* between them is far less affected than either absolute level. Any $W$-side level reported here is a lower bound across stable matchings and must never be compared against a $W$-proposing baseline without saying so.

### 3.4 The utility

Utility is additive over four terms: a status term, an attractiveness term, a distance penalty, and an indicator bonus for a status surplus. The bonus is what makes the model hypergamic rather than merely assortative. It rewards a partner at or above the agent's own status by a margin $\delta$, and is the model's operationalisation of the requirement that a partner rank at or above you rather than beside you.

Two parameters modify the utility's realised shape. A per-agent lognormal multiplier with coefficient of variation $\sigma_{\text{taste}}$ introduces taste heterogeneity, so different agents weight the same traits differently, which is the same device by which idiosyncratic tastes are added to a transferable-utility marriage model [Choo and Siow, 2006]. A persistent pairwise chemistry shock with standard deviation $\sigma_{\text{shock}}$, drawn once per ordered pair and held across epochs, enters in absolute utility units. Neither has an empirical anchor, and §5.8 treats both as knobs whose value changes the headline rather than as estimates.

### 3.5 The two environments

Environment A is the global market: a large population, visibility mediated globally by the learned ranker, no signalling cost, no entry filter.

Environment B is a subgraph of the *same* population. Agents are drawn into a pool of size at most $\phi N$, filtered on an observable value score, and within the pool visibility is complete, since every entrant sees every other, so B has information and no liquidity where A has liquidity and no information. The comparison is therefore a comparative static in an *institutional* parameter at a fixed population, which is a different exercise from the comparative statics in market size that the matching literature already treats [Crawford, 1991]; changing the subgraph size does move one side's size, and that channel is separated from the institution effect by holding the population fixed and varying only the shares. Proposing in B costs $c_B > 0$ and succeeds with probability $p$, so a proposal's presence carries information about the proposer's intent. A defection penalty $d$ is charged to a defector after a match dissolves.

The signalling mechanism is the model's core. An agent proposes only when $\tau_i V_{ij} p > c$, so the set of proposers is exactly $\{i : \tau_i > c/(V_{ij}p)\}$, and the separating threshold is

$$\tau^*(c) = \frac{c}{\bar{V}p}. \tag{1}$$

At $c \to 0$ the threshold vanishes, every type proposes, and a received proposal carries no information about intent: the posterior equals the population mean. This is a pooling outcome reached through a *cost* channel rather than through unobservable quality [Akerlof, 1970], and the paper treats it as an analogy to the classical lemons result rather than as an application of it. At $c > 0$ the threshold is positive, low-intent agents exit by their own incentive to avoid the cost, and the posterior rises: a separating outcome in the single-crossing sense [Spence, 1973]. A and B are therefore two settings of one parameter, not two theories.

An agent who cannot infer intent from a proposal must use something else, and the model's rational response is to raise the bar on observables. That is the acceptance threshold $\theta$, and its derivation from the pooling result rather than its assumption as an exogenous contagion is the content of H2.

### 3.6 Welfare, and the comparability rule

Welfare is defined per epoch over the whole population: the value created by matched pairs, less the signalling cost spent on proposals, less the loss from dissolved matches including the defection penalty.

The accounting rule is what makes the comparison meaningful. Agents excluded by B's entry filter are not deleted, and agents who fail to match inside B are not dropped: every agent appears in the welfare sum exactly once, and an agent B excludes is scored at whatever A would have given them. Without this rule an entry filter would look like a gain for the purely arithmetic reason that it removes the agents it cannot serve. This is a response to a known objection rather than a safeguard the project devised, since restricting choice can raise the welfare of those inside the restriction [Halaburda et al., 2018], and the model prices the cost to non-participants instead of ignoring it.

Both accountings are computed. The common one, over all $N$ agents, is the number the paper reports. The selected one, over entrants only, is computed so that the selection artefact can be reported honestly rather than hidden, and is never quoted as a headline.

### 3.7 The $\theta$ gate and its binding condition

The one feedback channel in the model runs from clearing outcomes to visibility. After each epoch, an acceptor whose received proposal count is informative updates $\theta$; $\theta$ is the quantile of her own utility distribution above which a candidate is admissible, so raising $\theta$ shortens her admissible set and lowering it lengthens it.

The gate binds only under a condition that is worth stating exactly, because several results below are regime-specific and the regime is set by it. Let $\phi$ be the target admissible fraction, the share of candidates an agent finds admissible at epoch 0, and let $k$ be the shortlist length. An agent's admissible set at epoch 0 has about $\phi N$ members. The gate can only truncate an agent's visibility if that set is *shorter* than the shortlist, since otherwise she takes the top $k$ of a larger pool wherever $\theta$ sits and the update moves nothing. So the gate binds when

$$\phi < \frac{2k}{N}. \tag{2}$$

At the shipped values $k = 20$ and $N = 1000$ this is $\phi < 0.04$. The config ships $\phi = 0.02$, inside the binding region; the earlier shipped value was $\phi = 0.2$, outside it, and the ablation data for that regime is labelled *inert* throughout. The factor of two in (2) reflects that the relevant comparison is against the candidate set on each agent's own side of the market rather than against the whole population.

Three consequences follow, and they are used later. First, $\phi$ and $k$ interact: raising the shortlist length or lowering the population requires lowering $\phi$ to keep the gate live, and the configuration validator warns when the rule is broken. Second, the difference between an inert run and a live run is a statement about whether the market responds to its own clearing outcomes, so an ablation measured at $\phi = 0.2$ is not a noisy version of one measured at 0.02; it is a different dynamical system. Third, the gate's liveness is a property of one epoch's calibration, and it relaxes: the update pulls $\theta$ toward a yield in the unit interval from an initial value far above it, so the gate truncates at the start and stops truncating within a few epochs. That relaxation is why the live and inert regimes produce nearly identical final-epoch gaps despite different trajectories.

## 4 Design

This section states what was run, at what sample size, against what controls, and what was held fixed, so a reader can tell which numbers in §5 are comparable to which and can re-derive any of them from the shipped configuration without re-running the simulation.

### 4.1 Regimes, and why the horizon is a design variable

The shipped configuration is $N = 1000$ agents, shortlist length $k = 20$, gate target $\phi = 0.02$, nominal horizon 100 epochs. Four regimes appear, and the differences between them are results in their own right:

| regime | $\phi$ | epochs | what it isolates |
|---|---|---|---|
| `inert20` | 0.2 | 20 | gate never binds; market deaf to its own outcomes |
| `live20` | 0.02 | 20 | gate binds, then relaxes |
| `live100` | 0.02 | 100 | the shipped horizon |
| `long300` | 0.02 | 300 | convergence, not a steady state |

The inert regime is retained rather than discarded because the difference between it and the live regime is the evidence that the feedback channel matters at all. The two 20-epoch regimes differ in $\phi$ alone; the two live regimes differ in horizon alone, and the first 100 epochs of the 300-epoch run are identical row for row to the 100-epoch run, which checks that horizon is the only change.

### 4.2 Arms

The A-side ablation compares four arms on a shared population, seed and utility set, so differences are attributable to the visibility rule rather than to sampling:

- **ncf**, the learned ranker. The treatment arm.
- **oracle**, which ranks by true utility under the same proposer direction and shortlist length. The benchmark, not a deployable institution.
- **popularity**, which ranks by aggregate engagement; its exposure Gini sits at the analytic closed form, which checks the exposure machinery.
- **random**, which ranks uniformly, because distance from uniformity is not a measure of harm and without it a reader cannot tell whether the ranker's concentration is unusual or what any non-personalised rule does.

The institutional comparison adds Environment B at four signalling costs ($c_B \in \{0.01, 0.05, 0.10, 0.25\}$) crossed with ten subgraph sizes ($\phi_B \in \{0.02, 0.05, 0.10, 0.15, 0.25, 0.35, 0.50, 0.65, 0.80, 1.00\}$): forty cells, five seeds, both accountings. Environment A there is the *oracle* arm, so nothing is estimated in either environment; the NCF-mediated comparison is reported separately and reverses the ordering. The drift decomposition is a $2 \times 2$ factorial over exogenous drift and algorithmic mediation, each on or off, from which the three terms are estimated.

### 4.3 Sampling and what counts as a replicate

Five seeds (42 through 46) underlie every headline, and this is the design decision most likely to be objected to, so the response is registered rather than deferred. A seed is a population, not a run: each draws fresh traits, intents, alignments and pairwise shocks and holds them fixed across arms, so arms are compared *paired* by seed and every gap is a per-seed difference.

A twenty-seed run (42 through 61) was executed at the shipped regime. A power calculation against the twenty-seed standard deviation requires fewer than five seeds to detect every headline effect at 80% power, with raw requirements below one in every case, and no headline changes sign or crosses zero at any seed count from five to twenty (`docs/STATS.md` §2.4). That is a statement about *sign*, not *width*. Two quantities behave badly on width: the five seeds are the minimum-variance contiguous block of the sixteen available for algorithmic loss and second-lowest for clearing, so their standard deviations understate the population's by roughly a factor of three and two and a half (§2.2). Every interval on those two is reported from the twenty-seed run, and no headline rests on a five-seed interval for either.

### 4.4 Intervals, multiplicity and the sweep

The estimating method is fixed before results are read and differs by seed count for a measured reason. At five seeds the Student-$t$ interval is quoted; at twenty, the BCa bootstrap with 10,000 resamples. The BCa interval is reported beside the $t$ interval at five but never quoted alone, because

Table 1: Validation against the closed-form results: the deficit identity and height-deficit form in the first five rows, assortative uniqueness and its precondition in the last two. These are the only numbers in the paper that do not trace to a file under `data/processed/`: they are asserted in `tests/python/test_validation.py` and transcribed in `docs/validation.md`, with no committed artefact between assertion and table. Every other table is generated from committed CSVs by `paper/tables.py`.

at that size its half-width is narrower than the sample's own standard deviation (`docs/STATS.md` §1). Bootstraps are seeded from a fixed constant and regenerate byte for byte.

Two families are corrected. The institutional grid holds 200 rows but 40 distinct cells (ten sizes by four costs), each replicated over five seeds; a cell is what was selected over, so its seeds are replicates rather than five further chances to win, and the family is 40. Uncorrected family-wise error under independence would be 0.871, but the cells are not independent, since adjacent sizes share populations and the same seeds are reused, so the realised rate is lower and is not claimed; the nominal figure is quoted as an upper bound. The decomposition corrects over its three terms.

Every committed result sits at one setting of four knobs, each swept at three levels with the others shipped: taste heterogeneity $\sigma_{\text{taste}} \in \{0.2, 0.5, 1.0\}$, chemistry noise $\sigma_{\text{shock}} \in \{0.5, 1.5, 3.0\}$, the weight ratio $\beta/\alpha \in \{1, 2, 3\}$ and the hypergamy margin $\delta \in \{0, 0.5, 1\}$: twelve cells, five seeds, 100 epochs. Four claims were fixed with their thresholds before the cells were read, so a reader can disagree with the thresholds rather than only with the verdicts: that the gap excludes zero and is positive (C1), that its point estimate is at most 0.05 (C2), that the ranker reaches at least 75% of agents (C3), and that clearing is non-degenerate (C4). C2's ceiling and C3's floor are the two judgments most exposed to the charge of being chosen after the fact, and the only mitigation is that they were written down first. The shipped cell appears in three axes with identical numbers (+0.025973, reach 0.9566), which checks that the axes are independent.

### 4.5 Validation against closed-form results

Before the recommender is asked anything, the matching and survival-function machinery is checked against proved results. This is verification, not validation: agreement with a theorem shows the code is correct and says nothing about whether the model describes a population.

Two tools are used because the results are of two kinds. The deficit results concern the largest matching under a *compatibility* constraint, which is a maximum-cardinality bipartite matching problem and is solved as one; a stable-matching procedure does not compute it. The uniqueness result is the stability question and is tested with deferred acceptance.

Table 1 reports both. The most informative check is the last: adding noise of order one percent of the status scale to each agent's own view of the other side leaves stability intact, with zero blocking pairs in both proposer directions, while uniqueness fails. The theorem's conclusion therefore rests on the lists being *common*, not merely correlated with status.

| Check | $N$ | Measured | Evidence | Verdict |
|---|---|---|---|---|
| Thm 2.2, deficit $D$ | $N = 300$ | exact on 5 seeds | 0.2433 / 0.3367 / 0.2200 / 0.2833 / 0.1800 | matching = empirical sup |
| Example 2.4 | $N = 20{,}000$ | 0.2519 vs 1/4 | single seed, tol 0.02 | agrees |
| Thm 3.3, $D(14.0)$ | $N = 5{,}000$ | 0 | exact | agrees by construction |
| Thm 3.3, $D(30.48)$ | $N = 5{,}000$ | 0.7254 | seed sd 0.0054 | formula 0.7217 |
| Thm 3.3, unequal var. | $N = 5{,}000$ | 0.7433 | seed sd 0.0053 | formula 0.7380 |
| Thm 5.1, unique | $N = 300$ | same matching both ways | 0 blocking pairs, 5 seeds | agrees |
| Thm 5.1, perturbed | $N = 300$ | 22 / 8 / 11 / 12 / 8 pairs differ | 1% status noise | uniqueness fails, stability holds |

What this licenses is narrow: it shows the deficit and stability machinery is correct, and does not show the recommender's unmatched mass has a closed form. The reasons are structural. The

Table 2: The hypothesis ledger: hypothesis, regime, point estimate, both 95% interval constructions, verdict. The $t$ and BCa columns are printed side by side so that the choice between them is visible rather than implied; the $t$ interval has nominal coverage at five seeds and is the one to read where they disagree. The H4 row alone carries a multiplicity correction (Bonferroni across 40 cells) and its verdict says so; the other rows are uncorrected and their families are stated in the text. "Best of 40 cells" is a count of surviving cells, not a width in `size_fraction`. Every row traces to `data/processed/rigor/` and to §5.3, §5.5 or §5.6. H1, H2 and H6 are absent because no interval is registered for them; see §6.

deficit is a supremum because the compatibility constraint is a nested upper set on a single scalar, and a top-$k$ list from an estimated multi-attribute utility has neither property.

### 4.6 What was not run

Three gaps are stated rather than left for discovery. The 300-epoch convergence run exists only at the shipped cell, so convergence is established at one point and not across the sweep, which is 100 epochs deep everywhere. The full-information benchmark materialises an $O(N^2)$ preference matrix and is capped at $N = 5000$, so larger-population results are NCF-only and lose their reference. And the model has no attrition mechanism, so one hypothesis cannot be tested at all (§5.7).

## 5 Results

The six hypotheses are reported one to a subsection. Each opens with the statement tested, gives the measured result with an interval where one is registered, and closes by naming what would have refuted it and whether that happened. The convergence result gets its own subsection because it asks about a limit rather than testing a hypothesis. Table 2 collects the four hypotheses with a committed interval; the other two have none, for reasons stated in place.

Two conventions hold. Every interval is a 95% interval over the seed distribution, and table 2 prints both the Student-$t$ and the BCa bounds for every row rather than choosing one silently, because which is preferred is a small-sample question and the two disagree at five seeds. The $t$ interval is the one with nominal coverage at n = 5; the BCa bootstrap attains roughly 84% there and returns a half-width narrower than the sample's own standard deviation (`docs/STATS.md` §1). At twenty seeds the two agree to 0.002. Where a row's interval has been widened for multiplicity, which is the H4 row only, the caption and the row both say so: the H4 bounds are Bonferroni-$t$ across the 40 institutional cells, and the uncorrected and Bonferroni-BCa bounds are beside them in the text (§5.5; `docs/STATS.md` §5.3). Gaps are always the paired per-seed difference, never a comparison of two arms' overlapping intervals.

| Hyp. | Regime | Measured | $t$ interval | BCa interval | Verdict |
|---|---|---|---|---|---|
| H3 | NCF − oracle Gini (live100, 5 seeds) | +0.0251 | [+0.0155, +0.0347] | [+0.0186, +0.0307] | survives; size is horizon-dependent |
| H3 | same gap (long300, 5 seeds) | +0.00044 | [-0.00068, +0.00155] | [-0.00029, +0.00105] | below 0.001 from about epoch 250 |
| H4 | welfare B − A (c=0.01, sf=0.65) | +64.62 | [+28.80, +100.43] | [+55.07, +78.46] | survives Bonferroni and Holm; best of 40 cells |
| H5 | interaction term (factorial, 5 seeds) | +16.70 | [-33.42, +66.82] | [-13.30, +53.05] | null under both; BCa includes zero too |

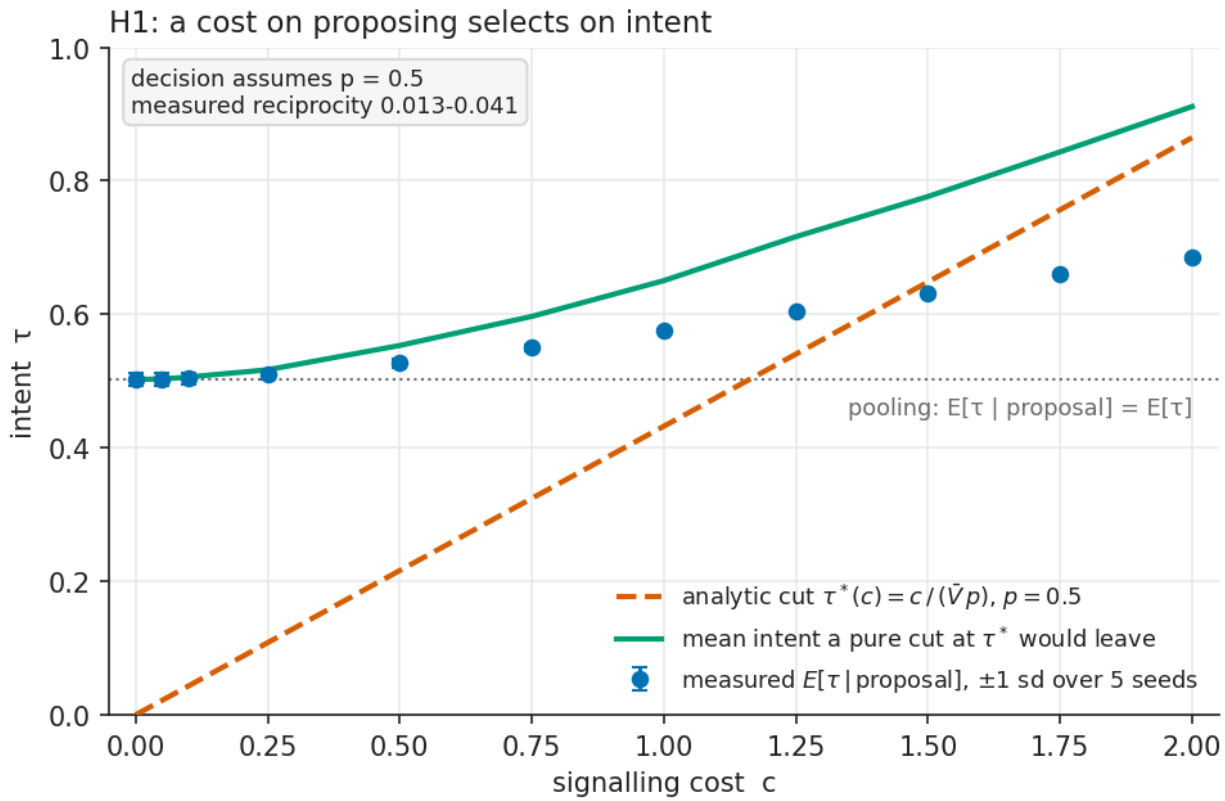



Figure 1: H1: measured mean intent of received proposals against signalling cost, with the analytic cut $\tau^*(c)$ and the intent a pure cut would produce. Earns its place as the only committed artefact for H1, and because it shows the discrepancy the text reports: the analytic curve assumes a reciprocity the market does not supply, so the measured points lie between the pooling line and the curve. No interval is registered for this quantity.

### 5.1 H1: a cost on proposing selects on intent

*Under test.* Mean received intent rises with the signalling cost $c$, and low-intent proposers exit. Kill condition: the mean intent is flat in $c$.

*Measured.* The direction holds: mean received intent is monotone increasing over the swept grid (Figure 1), so the kill condition is not met. The figure also shows the real finding. The analytic threshold $\tau^*(c) = c/(\bar{V}p)$ is evaluated at a prior success probability, while the reciprocity the market delivers is a small fraction of that prior. A threshold evaluated at a probability the market does not supply cuts more aggressively than the realised one, so the measured points sit between the pooling line and the analytic curve rather than on it. Selection happens, later and more weakly than the closed form predicts.

*Scope.* The hypothesis is testable only where the market stays open. Carrying measured rather than prior reciprocity, the market closes for good a short way above the bottom of the grid (`docs/THESIS.md`, H1 caveat; `docs/THEORY.md` §10.2), so the surviving statement is narrower than the one posed.

*Refuted?* No. Flatness is required and there is none. What is refuted is the implied magnitude. No interval is registered for this quantity.

### 5.2 H2: threshold inflation as defence against pooling

*Under test.* The acceptance threshold $\theta$ rises with received volume *net of* the mean intent of those proposals, and stops rising when $c$ makes the signal informative. Kill conditions: unrelated to volume, or still rising at high $c$.

*Measured: there is no instrument.* The shipped $\theta$ update reads one thing, the count of proposals received through clearing, and reads it only as a yield denominator. It never reads received intent. The count is bounded above by one per proposer while $\theta$ starts far above the unit interval a yield can occupy, so when the update fires it can only lower $\theta$, for a minority of acceptors, by a fixed step. No path exists by which a net-of-intent slope could be positive, because the intent term is not in the code. The observed decline of $\theta$ in volume is not evidence either way: it reflects which acceptors the update reaches, and holding reach fixed the association vanishes.

*Would it survive if the mechanism existed?* Partly, and self-defeatingly. Lengthening the shortlist scales volume but also moves the gate, so it is not a clean instrument. A flooding term added by hand does produce the predicted sign, but only at the largest weight tested and the smaller shortlist, and at that weight it raises the most-swiped acceptors' thresholds until reciprocity falls to zero and the market closes (`data/processed/flooding/`; `docs/THESIS.md`, H2). A defence against pooling that empties the market is not a defence.

*Refuted?* Formally yes, substantively uninformative: the mechanism is absent from the model rather than present and ineffective. Reporting this as a refutation would be a category error and as support would be worse. H2 is untestable under the shipped rule, and that is the finding.

### 5.3 H3: estimation error concentrates exposure

*Under test.* On a shared population and seed, the exposure Gini under the learned ranker exceeds that under an oracle ranking by true utility, because the ranker cannot represent every agent and those it cannot represent receive no exposure. Kill condition: the ranker matches or beats the oracle.

*Measured.* Not refuted. At the shipped regime and horizon the paired gap is $+0.0251$ with a $t$ interval of $[+0.0155, +0.0347]$, positive on all five seeds (`docs/STATS.md` §1, row 2; `intervals.csv`). The twenty-seed run gives $+0.0237$ $[+0.0204, +0.0271]$, which is the same quantity on a superset of the same seeds: seeds 42–46 are the first five of 42–61, so the two figures agreeing is expected and is not a test of anything. What it does establish is the interval's width, 0.0035 on the BCa construction at twenty seeds against 0.0121 at five, and that the point estimate moves by less than its own five-seed interval when the other fifteen seeds are added. A reader who wants the five-seed accident ruled out should read the sensitivity sweep below, where the same ordering is measured on twelve independently varied configurations, rather than the overlapping-sample agreement. The gap is positive with an interval excluding zero in all twelve sweep cells and reverses nowhere (§5.8).

*Refuted?* No in direction; the substantive version, that estimation error is a *standing* source of concentration, is refuted by §5.4. The inequality holds; the mechanism attributed to it does not.

*What it is not.* The oracle ranks by true utility under the same direction and shortlist length: an upper bound on what this architecture could achieve, not a deployable alternative. A gap against an oracle measures how far a learned ranker falls short, not what a better-designed platform would do.

### 5.4 The cold-start result, and its convergence test

*The question.* H3's gap is positive at every horizon measured. Is it a floor or a transient? A hundred-epoch run cannot answer, because the gap is still falling at epoch 99. The run was extended

to 300 epochs at the shipped configuration, five seeds (`data/processed/rigor/long_horizon/`; `docs/STATS.md` §3).

*Trajectory.* The mean gap rises to a peak shortly after the start, then decays monotonically in the reported points:

| epoch | 0 | 6 | 19 | 49 | 99 | 124 | 199 | 250–299 |
|---|---|---|---|---|---|---|---|---|
| gap | +0.2633 | +0.3055 | +0.2869 | +0.1718 | +0.0241 | +0.0072 | +0.0025 | +0.00037 |

The peak is at epoch 6. The hundred-epoch figure is not a floor: at epoch 99 the gap is roughly fifty times the residual it settles at, though both numbers are imprecise enough that the ratio is an order of magnitude, not a measured constant (`docs/STATS.md` §3.2).

Two cautions on reading the table. The points after epoch 124 are all within one final-epoch seed standard deviation of each other (0.0111), so the decay is established to about epoch 124 and the later entries are ordered noise rather than a measured continuation: the claim that the series is *monotone* holds only of the points printed, not of the underlying trajectory. And the entries at 124, 199 and 250–299 are small differences of quantities whose own intervals are wide relative to their distance from zero, which is why no ratio is quoted as a measurement.

*Convergence test.* The tail mean over epochs 250–299 is +0.00037 with a 95% $t$ interval of $[+0.000013, +0.000735]$, all five seeds positive ($t = 2.88$, $p = 0.045$; Wilcoxon $p = 0.063$). The BCa interval on the final epoch is $[-0.00029, +0.00105]$ and includes zero (`long_horizon_asymptote.csv`). The methods disagree about whether a residual exists and agree about its size, at most seven parts in ten thousand of the Gini scale.

*What "converges" is doing here, and what it is not.* The word appears elsewhere in this paper and carries more than the design delivers. A limit is not observed; a tail is. The measured statement is that the gap is below 0.001 from about epoch 250 and is no longer distinguishable from flat at five seeds. A flat tail is also what a very slow decay looks like, and no test here separates the two: the stationarity result below cannot, because a ten-epoch window at five seeds cannot detect a change of a size that would still matter. The paper therefore reports a bounded residual, not a convergence rate, and does not quote +0.00037 as a floor.

*Stationarity.* A paired test of epochs 290–299 against 280–289 gives +0.00010 ($p = 0.324$; Wilcoxon 0.313; three of five seeds higher). The largest single step in the last ten epochs is 0.00041, or 3.7% of the final-epoch seed standard deviation (`long_horizon_stationarity.csv`). The trajectory has stopped resolving above population noise, so 300 epochs suffices here and 100 does not. But "stops resolving" is a statement about the measurement, not about the process.

*The curve fit, reported and not trusted.* AIC and BIC prefer decay-to-constant over exponential decay decisively ($\Delta$BIC = 23.9 from the peak), but the constant it estimates is *negative* with an interval excluding zero while the observed tail is positive: a single exponential cannot represent a series that first rises, so the constant absorbs the misfit rather than estimating an asymptote (`docs/STATS.md` §3.1; `long_horizon_decay_fits.csv`). The comparison answers which curve fits better, not what the asymptote is.

*What this establishes.* That the exposure gap at the shipped configuration is a cold-start property that disappears, and that a hundred-epoch figure describes a transient. Not that the mechanism is the one H3 offers (§5.8 finds a cell where coverage never climbs and the gap closes anyway), nor that convergence holds elsewhere, since the 300-epoch run exists only at the shipped cell.

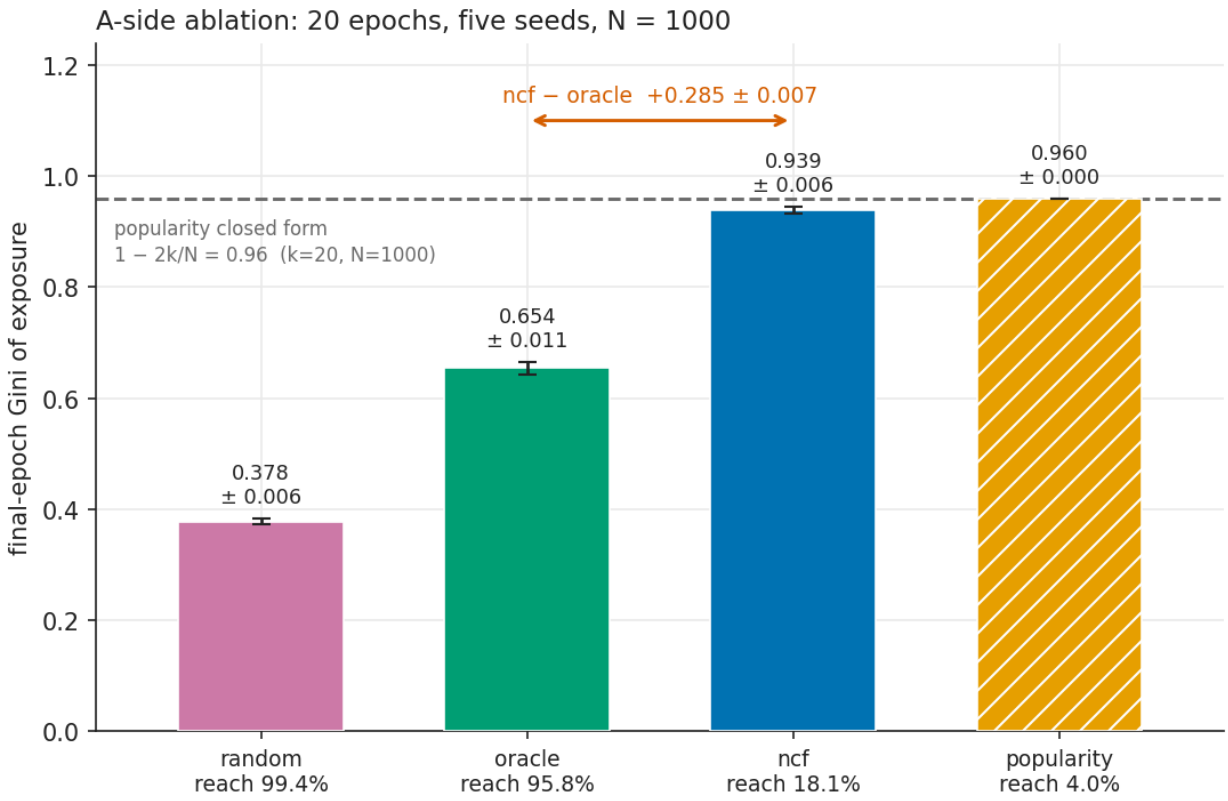



Figure 2: The four arms at the end of a 20-epoch ablation, five seeds. Earns its place as the only committed artefact showing the H3 inequality against both controls at once: a random baseline, which measures distance from uniformity rather than harm, and an oracle, which is unattainable. The popularity arm sits at the analytic closed form for a popularity ranker, checking that the exposure machinery behaves as designed; the reach labels are the mechanism's own diagnostic and are what §5.8 interrogates. Source: `data/processed/ablation/`.

### 5.5 H4: an interior optimum in the institutional comparison

*Under test.* Welfare under Environment B exceeds welfare under A for subgraph sizes above a critical mass and falls below it for small sizes, at fixed cost. Kill condition: the ordering is monotone in size: B always wins, so the thin-market penalty does nothing, or always loses (the information gain does nothing). Welfare is scored on a common basis, with agents B excludes priced at their counterfactual A outcome.

*Measured.* Survives on one slice, killed on the others. At $c_B = 0.01$ against an oracle A, the optimum is size_fraction 0.65 with a residual welfare difference of +64.62, $t$ interval [+52.37, +76.86] (`docs/STATS.md` §1, row 7b), and the ordering there is non-monotone, which is the shape H4 asks for (Figure 3).

*The selection correction the claim requires.* The optimum is the maximum of forty cells, so an uncorrected interval on it is not an interval on anything. Corrected over that family, the Benjamini–Hochberg $q$-value is $3.9 \times 10^{-4}$, Holm rejects, and the Bonferroni $t$ interval widens to [+28.80, +100.43] (`docs/STATS.md` §5.3; `multiple_comparisons.csv`). At roughly eleven standard errors the correction is cheap.

*The narrower finding.* Of seven cells where B beats A, only two survive correction (0.65 and

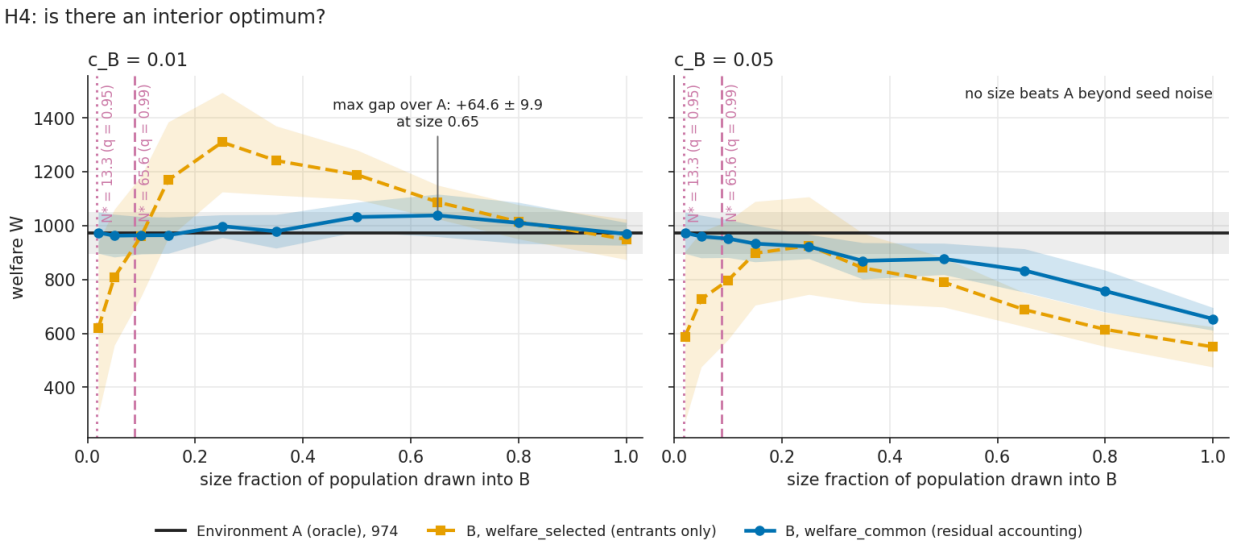



Figure 3: H4: welfare against the share of the population drawn into Environment B, at two signalling costs, with Environment A as a horizontal line. Earns its place because one frame carries all three elements of the verdict: the non-monotone shape at the low cost, the monotone collapse at the high cost where no size beats A, and the gap between selected and common accounting (dashed against solid), which is the selection artefact the comparability rule exists to price. Vertical rules mark analytic critical masses; they serve the thin-market penalty and do not explain the right-hand slope, which `data/processed/THIN_MARKET_README.md` attributes to top-$k$ truncation.

0.50). The optimum is a two-cell feature, not a region; the curve beyond those cells is indistinguishable from zero. A caution the correction does not address: the optimum's standard deviation is the ninth smallest of the forty, at a little under a third of the median cell's spread, so it is a cell where five seeds happened to agree, which is why the $t$ interval is the one to read.

*What the correction does not fix.* Bonferroni and Holm widen an interval for having searched a family; they do not undo the fact that the cell was selected on low variance as well as high mean. The interval above is therefore an honest width around a point that may be biased upward by the selection itself, and the two are not the same problem. The clean answer is to re-estimate the optimum on seeds that took no part in choosing it, which this study does not have: the twenty-seed run covers seeds 42–61 and the optimum was chosen from the same five that the rest of the paper uses. The per-cell seed sd spans a wide range (`intervals.csv`, `welfare_diff_residual_within_cell_sd`), so a reader should treat +64.62 as an upper estimate of the interior gain rather than a central one.

*Killed elsewhere.* At $c_B \geq 0.05$ B always loses: no size beats A beyond seed noise (`data/processed/institutional/`). Against an NCF-mediated A rather than an oracle A the ordering reverses and B always wins, so the result is about the quality of A's visibility, not subgraph size as such.

*Refuted?* Yes, on the $c_B \geq 0.05$ slice, and yes in that the hypothesis is a property of one cost level rather than of the model. What survives: at a cost low enough that entry screens little, an interior optimum exists and is two cells wide.

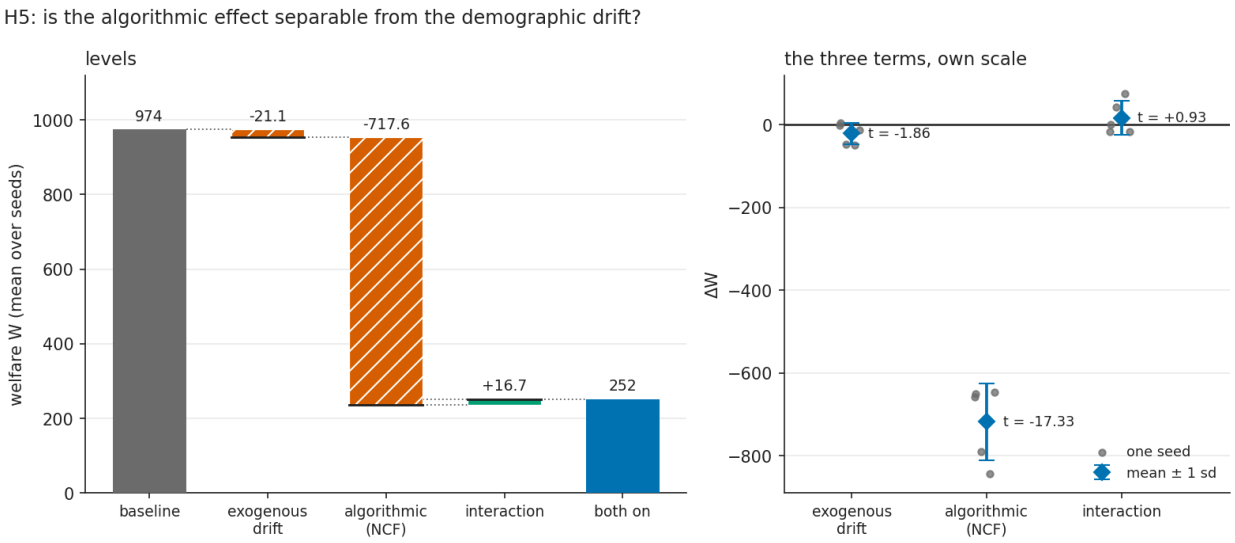



Figure 4: H5: the four factorial levels and the three decomposed terms. Earns its place because the left panel shows the decomposition's arithmetic, with the interaction as the residual closing the gap between the marginal effects and the joint effect, while the right panel puts the three terms on their own scale with per-seed points, which is what makes the scale difference between the algorithmic term and the other two visible. Without the left panel the terms cannot be checked to add up; without the right panel the interaction looks larger than it is. Source: `data/processed/institutional/factorial.csv`.

### 5.6 H5: the decomposition's interaction term

*Under test.* The welfare change from an exogenous status shift and from algorithmic mediation decomposes into additive and interaction parts, with the interaction estimated rather than assumed. The hypothesis does not predict its sign; it predicts the term is estimable and may be near zero.

*Measured.* The interaction is $+16.70$ with a $t$ interval of $[-33.42, +66.82]$ and a BCa interval of $[-13.30, +53.05]$; its BH $q$-value over the three terms is 0.407 and it is not rejected (`docs/STATS.md` §5.4). Indistinguishable from zero, as H5 allowed.

*The terms that are not zero.* The algorithmic term is $-717.61$ $[-795.77, -651.37]$ and dominates: mediation costs an order of magnitude more welfare than the demographic shift moves it. The exogenous term is $-21.10$, and it is where the two methods disagree: its BCa interval $[-41.13, -3.42]$ excludes zero while its $t$ interval $[-52.53, +10.32]$ and $q$-value of 0.204 do not. At five seeds with this skew the BCa interval is optimistic, so the defensible reading is a negative sign with unresolved magnitude. Figure 4 shows the arithmetic and the scale difference.

*Refuted?* No. The kill condition was incomparability of the factorial cells, which would leave the decomposition undefined; comparability is asserted per seed and holds. The measurement establishes a negative result reported honestly: drift and mediation act through separate channels here, and the claim that the algorithm amplifies the shift has no support.

### 5.7 H6: differential attrition by intent

*Under test.* High-intent agents leave Environment A at a higher rate than low-intent agents. Kill condition: attrition flat in $\tau$, or the sign reversed.

*Measured: not at all.* The model has no attrition mechanism. Agents do not exit in response

Table 3: The sensitivity grid: one knob at one level per row, the rest at shipped values, five seeds, 100 epochs. Gap is the paired exposure difference and the interval its BCa 95% bounds. The shipped level of each knob is italicised and appears in three axes with identical numbers, which checks that the axes are independent. C1–C4 are the claims of `docs/STATS.md` §4.1, bold where a claim fails.

to outcomes; the only population change is generational turnover, which is not a decision and does not read $\tau$. No committed artefact exists for this hypothesis and no quantity is registered, so no number is quoted.

*What can be said without one.* The sign of any measured correlation is determined by the exit rule imposed, and the two examined point opposite ways: keying exit on being unmatched yields no association, since matching does not depend on $\tau$; keying it on net payoff yields a strong positive one, but by cost burn rather than rejection: high-intent agents propose far more often and are matched at the same rate (`docs/THESIS.md`, H6). Neither tests the hypothesis, which is about a market with an exit margin.

*Refuted?* Neither confirmed nor refuted: untested, and reported as such. The cost-burn channel it exposes is the same one that kills H4 at high signalling cost.

### 5.8 Robustness across the parameter sweep

The hypotheses above were each measured at one setting of four modelling knobs. This subsection asks where they hold, which is why the paper can report a size for the exposure gap at all, since one cell cannot separate a property of the mechanism from a property of its corner. Twelve cells, one knob moving at a time, four claims fixed with their thresholds before the cells were read (Table 3).

| Knob | Level | Gap | 95% BCa | C1 | C2 | C3 | C4 |
|---|---|---|---|---|---|---|---|
| taste/sigma | 0.2 | +0.0268 | [+0.0209, +0.0326] | yes | yes | yes | yes |
| | *0.5* | +0.0260 | [+0.0164, +0.0348] | yes | yes | yes | yes |
| | 1.0 | +0.0265 | [+0.0184, +0.0330] | yes | yes | yes | yes |
| shock/scale | 0.5 | +0.0097 | [+0.0062, +0.0116] | yes | yes | **no** | yes |
| | *1.5* | +0.0260 | [+0.0164, +0.0348] | yes | yes | yes | yes |
| | 3.0 | +0.0470 | [+0.0192, +0.0665] | yes | yes | yes | yes |
| beta/alpha | 1.0 | +0.0340 | [+0.0248, +0.0433] | yes | yes | yes | yes |
| | *2.0* | +0.0260 | [+0.0164, +0.0348] | yes | yes | yes | yes |
| | 3.0 | +0.0224 | [+0.0194, +0.0243] | yes | yes | yes | yes |
| delta | 0.0 | +0.0229 | [+0.0157, +0.0291] | yes | yes | yes | yes |
| | *0.5* | +0.0260 | [+0.0164, +0.0348] | yes | yes | yes | yes |
| | 1.0 | +0.0263 | [+0.0236, +0.0301] | yes | yes | yes | yes |

*Holds everywhere.* The H3 ordering holds in all twelve cells, positive with an interval excluding zero, reversing nowhere (C1). The gap is small by the pre-fixed ceiling in all twelve (C2), though the largest cell's upper bound reaches +0.0665 and is the weakest case. Clearing is non-degenerate in all twelve (C4). Neither the ordering nor the existence of a clearing market is an artefact of the shipped corner.

*What does not.* One cell fails the reach claim: at the lowest chemistry noise the ranker reaches

0.4616 of agents against 0.9566 shipped (C3). This is the mechanism switching off, not a small deviation: at low chemistry noise the filter never learns coverage, so the process §5.4 identifies as the reason the gap closes is unavailable. The gap is still small and positive there (+0.0097), but because *both* arms concentrate and the ranker has not caught up.

This is a conclusion reversing. The cold-start explanation is conditional: it requires the filter to learn whom it can represent. The convergence in §5.4 is therefore a fact about the shipped configuration, not a general property of the mechanism, and whether the gap converges at the cell where reach fails is untested.

*How large.* The gap ranges from +0.0097 to +0.0470, a factor of about five, with no change in sign; the shipped cell sits mid-range. Quote the range with its condition, or the shipped cell with its regime, never +0.025 as "the" gap.

*A documented condition that does not reproduce.* The configuration states, and a guard test pins, that the hypergamy margin $\delta$ has power only when $\beta/\alpha \geq 2$. Re-measuring shortlist overlap at the shipped heterogeneity and noise reverses the ordering: $\beta/\alpha = 1$ gives $\delta$ *more* power than 3, and all three levels clear the threshold the guardrail treats as meaningful (`docs/STATS.md` §4.2; `sensitivity/delta_power.csv`). The guard test is not wrong; it was measured on a noise-free homogeneous population, and the rule of thumb describes that population rather than the one the experiments run on.

What the sweep does show is that $\delta$ moves no headline: the gap is +0.0229 at 0, +0.0260 at 0.5 and +0.0263 at 1.0, a spread of 0.0034 against intervals about 0.012 wide, while the weight ratio moves it twice as far and chemistry noise four times as far. The defensible statement is that no claim here rests on $\delta$ at any weight ratio tested, not that the results require $\beta/\alpha \geq 2$.

*Refuted?* The sweep refutes the unconditional form of the cold-start explanation and the stated $\beta/\alpha$ condition. It does not refute H3's ordering, the one result that survives every cell.

# 6 Limitations

Ordered by how much each constrains what the results can mean, not by how easy it is to fix.

**Every parameter is a modelling choice.** Nothing is calibrated to data: the trait distributions are drawn for tractability, the taste and chemistry parameters have no empirical anchor, and neither the signalling cost, the defection penalty, nor the entry filter is estimated. The results bound what this mechanism *can* do under a stated utility and say nothing about whether any population sits at these parameters. Because traits are $z$-scored before entering a utility, only rank order matters, but a distribution with no variance would still destroy its term, so the defaults are deliberately not self-report-shaped and the attractiveness result is an upper bound on its influence rather than an estimate. The mitigation is that the two most consequential knobs are reported as a function of their value. It is partial: a sweep over four knobs is not a calibration.

**Five seeds carry every headline.** The power analysis says this is adequate for sign and inadequate for width on two quantities, and both matter. An interval on a model's seed-averaged quantity is not an interval on the model's answer. The five seeds were also not chosen for this purpose; they predate the twenty-seed run, which shows them to be an unusually low-variance block for two of the seven tracked quantities (`docs/STATS.md` §2.2). That finding is reported and is not repaired: the interval arithmetic every table uses was changed to the construction with nominal small-sample coverage, but the seeds themselves were left alone, so any claim whose variance happens to be underestimated by this particular block still reads as tighter than it is. The honest reading is that the two affected quantities, algorithmic loss and clearing rate, carry no headline here, and that a future study should choose seeds for the purpose rather than inherit

them.

The same caution applies to the H4 optimum without the review's help being needed: it was selected from forty cells on mean *and* low variance, and Bonferroni corrects the first without undoing the second (§5.5).

**The thread count is part of the run's identity, and that was found late.** Torch's intra-op reductions are not bitwise associative, so the same seed at one thread and at eight gives different numbers; across a hundred epochs of this feedback loop the divergence reaches 0.006–0.015 in exposure Gini, against a between-seed standard deviation of 0.011. Every configuration now pins `ncf.num_threads` to 1 and the contract is tested, but the reproducibility guarantee this project claims is a guarantee at a named thread count, not an unconditional one (`docs/INTERFACES.md` §3; `data/processed/rigor/thread_sensitivity.csv`).

**One population size carries the headline.** $N = 1000$ is the only size at which the full comparison was run, because the full-information benchmark needs an $O(N^2)$ preference matrix and is capped at $N = 5000$. Clearing rate is not comparable across population sizes at fixed $k$: it falls roughly as $1/N$, and scaling $k$ to compensate makes the swipe log grow superlinearly. The clearing figures are properties of $N = 1000$ at $k = 20$ and do not transport.

**There is no closed loop through preferences.** The ranker changes who is visible, not what anyone wants: the preference generator takes no shown-set argument. A reader looking for the classic feedback story will not find it here. A different loop exists, through the acceptance threshold, and it is live at the shipped configuration; but it relaxes within a few epochs, so even the loop that is present is a transient. This is the largest gap between this model and the recommender literature it engages, and it is a structural absence rather than a measurement problem.

**There is no attrition.** Nobody joins or leaves in response to outcomes, so there is no selection at the extensive margin and a market that deteriorated enough to empty out could not show it. This is why one hypothesis is untested (§5.7), and it bounds the welfare reading: a mechanism that repelled participants would look free.

**The defection penalty prices defection without deterring it.** It is charged after a match dissolves; no agent proposes differently, sets a different threshold, or stays longer because of it. Making it deter would mean feeding it into the proposal decision, which is a different model whose institutional result has not been measured.

**The benchmark is not an achievable institution.** The oracle ranks by true utility, so every gap measured against it is a distance from an unattainable bound rather than a choice an agent could make. Conversely, the institutional comparison scores B against an oracle A, and against an NCF-mediated A the ordering reverses. Both facts bound what the A/B result means: it is about the quality of A's visibility, not institutional form in general.

**Truncation is imposed, not strategic.** Agents receive their top-$k$ list and do not game it, so the strategic theory of truncation does not apply directly, and the model cannot say what happens if agents report their lists instead.

**Validation establishes verification, not validity.** Agreement with closed-form results shows the machinery is correct; it establishes nothing about the recommender, because the deficit those theorems characterise is measured under a compatibility constraint on a single scalar, which is exactly what makes it tractable, while this model's constraint is visibility through an estimated multi-attribute utility. A matching deficit is not evidence that the recommender's unmatched mass has a closed form. It does not appear to have one.

**Nothing here is a claim about people.** The intent type has no instrument, the utility has no estimated weights, and the environments are model constructs. Neither is ethically superior: B carries an entry filter that excludes agents, and the model prices that exclusion rather than endorsing it.

# 7 Conclusion

A learned ranker between the two sides of a matching market concentrates exposure relative to an oracle, and the concentration is a start-up cost rather than a standing feature. That is the paper's finding, and it is a finding about *when* the measurement is taken as much as about what is measured: at the shipped hundred-epoch horizon the gap is +0.0251 and looks like a floor; at three hundred epochs it is below 0.001 and no longer distinguishable from flat. The hundred-epoch figure is therefore an order of magnitude away from where the series settles, though the ratio between the two is not itself a measured quantity. A literature that reports single-horizon comparisons cannot distinguish a transient from a bias, and this mechanism supplies a case where the distinction is decisive. What is not established is the limit: a flat tail is compatible with a very slow decay, and separating them needs a horizon this study did not run.

Three qualifications travel with it. The ordering holds in every cell of a twelve-cell sweep, positive each time, but its size varies fivefold, so no single number is "the" gap. The explanation is conditional: the filter learning coverage accounts for the decay at the shipped setting and cannot account for it where coverage never climbs, and convergence has been tested at only one point in parameter space. And the institutional comparison yields an interior optimum that survives multiplicity correction but is two cells wide, so it is a location with narrow support rather than a curve with a maximum.

Two hypotheses did not survive as stated and one could not be tested. The mechanism the second names is absent from the shipped rule, and the model has no attrition margin for the sixth. Those are reported as findings rather than omissions, because a six-for-six ledger would be evidence of a design that could not fail.

What would move this from a model result toward a claim about markets is calibration, and the paper does not attempt it. Until then the results bound what this mechanism can do under a stated utility, and the bound is the contribution.

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